DATASET & PLATFORM REPORT

A Scalable Big Data Platform for Real-Time Urban Air Quality Tracking and Public Safety Alerting

23AID302 — Big Data Analytics

Structured, Semi-Structured and Unified Data · Model · Live Alerting · Serving Layer

Report date: 12 September 2026 · every figure is from a real, logged run

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1. Executive Summary

This report covers the project's complete data estate and everything built on it: a 27.48 GB structured dataset, two genuinely semi-structured layers pulled live from a public atmospheric-composition API, a 154.6-million-row unified table, a full-scale gradient-boosted forecaster, a working streaming alert engine consuming live data, and a Hive serving layer. Every number below is from a real run whose log is preserved under docs/run_logs/.

Headline model result (real rows only, n = 2,074,569): GBT R^2 0.8910, RMSE 21.806 µg/m³ against a persistence baseline of R^2 0.8587, RMSE 24.831 — a 12.18% error reduction, on an unsampled 18.7-million-row test window across 241 cities. The model passes the project's own gate by +0.0323 R^2.

Headline live result: 16,560 live city-hours enriched on YARN and scored by the streaming layer produced 1,130 alerts above the AQI 150 threshold, at roughly 7 seconds per micro-batch.

All five V's are now demonstrated, not claimed. Velocity and Value were the two that had been architecture rather than evidence at the start of this work block; both were executed on the cluster during it.

Equally important is what this report does NOT claim. The synthetic fraction of the structured layer (88.9%) is disclosed. Cross-validation against an independent source shows moderate agreement on particulates and essentially none on SO_2. Seven pipeline defects were found only by running code that had previously been written but never executed; each is listed with how it was caught. A request for 2010–2023 semi-structured coverage was declined because measurement showed the source returns ten years of empty rows for that span.

2. The Data Estate at a Glance

Property	Value
Root	F:\SEM 5 NOTES\Big Data\Project_Final\airquality_expanded
Structured layer	27.48 GB CSV, 146,276,379 rows, 32 columns, 2010-2023
Semi-structured - historical	8,263,440 records, 458 MB JSON, 2022-08 to 2026-09, 230 cities
Semi-structured - live	44,160 records, 6.7 MB JSON, rolling window, 230 cities
Unified table	154,583,979 rows, 4.3 GB Parquet, one 17-column schema
Total on disk	33 GB
In HDFS	raw + clean (10 GB) + features (19.7 GB) + unified (4.2 GB) + models, all fsck HEALTHY
Cities / states	241 / 31 (230 in the semi-structured layers)
Stations	4,077 = 453 real + 3,624 synthetic
Real vs synthetic (structured)	16,252,931 (11.1%) / 130,023,448 (88.9%)

3. Structured Layer (27.48 GB)

3.1 Origin

Rebuilt from a 23.44 GB / 48.6M-row / 139-column CPCB source covering 453 real stations. The rebuild keeps all 16,252,931 real rows verbatim and generates 8 synthetic sibling stations per real station with an AR(1)

autocorrelated perturbation (per-station φ, σ = 0.08), copying each real station's missingness pattern rather than densifying it. Result: 146,276,379 rows, 4,077 stations, 241 cities, 2010–2023, 23 CSV parts.

3.2 Measured fill rates and ranges

Stratified sample of 1,094,195 rows across all 23 parts (scripts/_audit/scan_expanded_fillrate.py):

Column	Fill %	Min	Max	Mean	Median
pm25	73.64	0.01	1237.42	68.91	45.76
pm10	65.13	0.01	1220.27	132.97	97.89
no2	78.44	0.01	623.40	28.89	19.39
so2	74.85	0.01	252.50	12.92	8.04
co	73.94	0.00	60.17	1.02	0.69
o3	72.98	0.01	265.36	30.67	20.37
nh3	58.59	0.01	618.73	25.45	17.86
solar_radiation	59.83	0.00	1654.57*	157.76	48.90
pressure	45.90	668.45	1147.29*	831.29	753.38
temperature_c	100.00	-6.16	58.79	25.20	24.29
humidity_percent	100.00	10.55	100.00	65.68	65.07
traffic_index	100.00	3.60	100.00	59.63	59.45
is_rush_hour	100.00	0	1	0.32	0

* *solar_radiation (max 1654.57 W/m²) and pressure (max 1147.29 mmHg) hold physically impossible outliers in the raw layer; both are nulled by the Phase 3 range-guard before imputation.*

3.3 Four defects found in the source, and repaired

#	Defect	Scale	Fix
1	is_rush_hour / variation_id not binary	24,725 / 29,918 distinct values	round, cast to int
2	T+1 label via positional lead()	8.69% of labels had the wrong horizon	require next_ts == ts + 3600 s
3	physically impossible values	humidity 104.999%, traffic_index 110.2	clamp to [0,100]
4	sparse simultaneous AQI coverage	43.18% carry all 7 pollutants, 15.72% carry none	three-tier median imputation, flagged

Defect 2 is the consequential one: a positional lead() on a non-unique key mislabelled roughly one training target in twelve with the wrong forecast horizon. It, not a modelling limit, was the root cause of an earlier run that scored R^2 = 0.58 — below the do-nothing persistence baseline.

3.4 Real vs synthetic

The synthetic tail is thinner than the real one: pm25 p99 366 vs 449 µg/m³, standard deviation 74 vs 99. Because the alert threshold is a tail event, synthetic rows under-represent exactly the regime alerting exists to catch. All model results are therefore reported on real rows separately, and that is the figure quoted.

4. Semi-Structured Layers (CAMS via Open-Meteo)

4.1 Source

Open-Meteo's Air Quality API, backed by Copernicus CAMS atmospheric-composition analysis (ECMWF). Free, public, no API key, no account. Every response is stored byte-for-byte as returned: nested hourly struct-of-arrays with its own hourly_units schema. CAMS is a modelled reanalysis product assimilating satellite and ground observations — real, widely used, but model output rather than raw sensor readings, and described as such throughout.

4.2 Why coverage starts in August 2022 and not 2010

The original request was for 2010–2023 to match the structured layer. That was tested rather than assumed (scripts/_audit/cams_history_availability.py, cams_history_cutoff.py):

Date range requested	What the API returned
before 2013-01-01	rejected: 'out of allowed range'
2013-01 to 2022-07	timestamps present, EVERY value null (0% fill)
2022-08	89.7% fill - real data begins
2022-09 onward	~99.8% fill (ammonia 0%)

Requesting the full 2010–2023 span would have produced ten years of rows that look like coverage and contain nothing. Coverage therefore begins at the first month carrying real values.

4.3 Two layers, and why both are kept

- Historical: 230 cities × 35,928 hours = 8,263,440 records, 2022-08-01 → 2026-09-05, 458 MB, fetched in 37.9 minutes. Fill 99.79% on all pollutants except ammonia (0%, a real CAMS regional gap, left null rather than backfilled).
- Live: 230 cities × 192 hours = 44,160 records, rolling window. Refreshed by the daily cycle.

They overlap by 192 hours per city, and the overlap is not redundant: for Delhi, 173 of those hours match exactly and 19 differ, all at the tail of the live window. That is CAMS revising forecast values into analysis values. Both are kept and tagged (live_api vs cams_history), because the forecast values are what the alerting layer actually sees in real time.

230 of 241 cities resolved. The other 11 (Bileipada, Byrnihat, Chikkaballapur, Davanagere, Eloor, Gummidipoondi, Kunjemura, Mandi Gobindgarh, Palkalaiperur, Sivasagar, Vatva) returned a genuine geocoder miss and were excluded rather than given invented coordinates.

5. The Unified Table (154,583,979 rows)

Built by scripts/01c_merge_structured_semistructured.py: Spark reads the structured CSVs and the nested JSON with their native readers — genuine schema-on-read across two data models in one job — projects both onto one 17-

column schema, and unions them. Values are never blended or imputed across sources; a row is tagged by where it came from.

source_format	data_source	rows	time range	cities
structured_csv	synthetic	130,023,448	2010-01 to 2023-03	241
structured_csv	real	16,252,931	2010-01 to 2023-03	241
semistructured_json	cams_history	8,263,440	2022-08 to 2026-09	230
semistructured_json	live_api	44,160	rolling 7-8 days	230
TOTAL		154,583,979		

Verified by a fresh Spark read-back, not the write-time log (scripts/_audit/verify_unified_dataset.py): total reconciles exactly against independently known counts; zero duplicate (city, timestamp) keys; every semi-structured city has exactly its expected row count; nulls appear only where the source genuinely lacks the field. That reconciliation check is what caught defect 1 in §9.

6. Cross-Validation: CPCB vs CAMS (Aug 2022 – Mar 2023)

The eight-month overlap is the only window in which two independent sources describe the same cities over the same hours. Structured rows were averaged to a city-hour (CAMS gives one point per city; CPCB gives many stations), using measured values only, and joined on (city, timestamp): 1,152,707 matched city-hours.

Pollutant	corr	bias	RMSE	mean CPCB	mean CAMS
pm25	0.565	+8.55	50.75	59.96	51.41
pm10	0.558	+44.89	97.18	121.82	76.93
o3	0.361	-48.54	67.17	28.54	77.07
no2	0.219	-2.07	35.24	24.87	26.94
so2	0.083	-10.05	43.32	14.60	24.65

$PM_{2.5}$ and PM_{10} track moderately, which is plausible for roadside sensors against a coarse model grid; a constant bias with decent correlation is the expected shape, not a failure. SO_2 shows essentially no agreement, and O_3 carries a large systematic offset. Per-city spread is wide: Kolkata 0.792 and Howrah 0.794 at the top, Pune −0.209 at the bottom. CO was excluded from the headline because the two sources report it in different units (mg/m³ vs µg/m³). A timezone check across hour offsets −3…+3 peaks at 0h/+1h and decays symmetrically, so the IST assumption holds.

One result was invalidated. A synthetic-vs-CAMS comparison returned figures numerically identical to real-vs-CAMS (0.564 vs 0.565), because averaging to city-hour means averages away the AR(1) perturbation. That test cannot distinguish the two by construction and is not evidence either way; generator fidelity is tested at station level by 00d_validate_expansion.py instead.

7. Model: Full-Scale GBT

Phase 4 at full scale (FeatureExtraction.scala, 73.5 min) reproduced this project's RUNBOOK to the row: 146,276,379 in → 128,660,967 after dedupe → 93,216,105 with a valid T+1 label → 100% assemblable, 86-dimensional vector, 19.7 GB. Training (03_train_gbt.py, 300 trees, depth 6, delta target, chronological split) ran 5h50m without the StackOverflow that killed the previous full-scale attempt at 3h50m.

	Track A (19 cities, 120 trees)	Track B (241 cities, 300 trees)
training rows	3,477,690	10,000,552 (seeded sample of 74.5M)
test rows	871,488	18,671,121 (not sampled)
real test rows	96,832	2,074,569
feature vector	78 dims	86 dims
persistence R2 (real)	0.8714	0.8587
GBT R2 (real)	0.8855	0.8910
persistence RMSE (real)	21.665	24.831
GBT RMSE (real)	20.436	21.806
RMSE reduction	5.67%	12.18%
gate	PASS (+0.0142 R2)	PASS (+0.0323 R2)

Evaluation is not sampled: all 18,671,121 test rows are scored, so the metric stands even though training used a seeded 10M-row sample of the 74.5M available (a RAM limit on a 15.7 GB host, not a pipeline limit). The 12.18% reduction finally substantiates the ~11.3% the original documentation quoted from an sklearn reproduction — as a real distributed Spark GBT on the full data.

The eight extra vector dimensions between Track A and Track B are the climate types the 19-city model never saw; 54 of 241 cities had been landing in the encoder's catch-all bucket. Worst cities by RMSE (Kochi 100.7, Pune 98.4, Byrnihat 78.7) are the same ones the cross-validation flagged as poorly correlated.

8. Live Alerting: Streaming + Phase 8b

8.1 What the streaming layer does

AirQualityStream.scala watches an HDFS directory with a 10-second trigger. Each micro-batch rebuilds the Phase 4 features (using a persisted 30-hour history table so lag features survive restarts), loads the GBT, predicts the next-hour $PM_{2.5}$ delta, converts to CPCB AQI, and writes an alert with severity and dominant pollutant wherever forecast AQI exceeds 150.

Batch	rows in	predicted	alerts	latency
0 (replay)	120	15	10	10,750 ms
1 (replay)	120	15	10	7,817 ms
2 (replay)	120	14	10	7,019 ms
3 (replay)	120	12	7	6,894 ms
LIVE (Phase 8b)	16,560	16,560	1,130	23,452 ms

8.2 The gap Phase 8b closes

The first streaming run was fed rows already in the structured 32-column schema. Raw CAMS JSON supplies only 28 of the model's 54 numeric features:

Missing from raw CAMS JSON	Count	Real source used
weather: temperature, humidity, wind speed/direction, rain, pressure, solar	8	Open-Meteo weather API (same provider, free, no key)
derived from weather: ventilation, scrub, stagnation, temp x humidity, is_raining	5	computed once weather exists
traffic_index, is_rush_hour, urbanization_index	3	city_profiles.csv - the same profiles the generator used
*_was_imputed provenance flags	10	0 where the API returned a value, 1 where it returned null

LiveEnrichment.scala (Spark, submitted to YARN as application_1788670113534_0002) joins CAMS JSON, weather JSON and city_profiles.csv into the schema the stream consumes. Two unit conversions were measured, not assumed: CO µg/m³ → mg/m³ (÷1000; unconverted, structured CO averages 0.95 against CAMS 566), and pressure hPa → mmHg (×0.750062; output 753.81 matches the structured median of 753). Wind is requested from the API in m/s directly.

Result: 16,560 live rows across 230 cities, zero nulls on model-critical columns, all 16,560 predicted, 1,130 alerts. Example: "Air quality warning for Jaisalmer: forecast AQI 500 (Severe), driven by pm10. Forecast PM2.5 216.9 µg/m³ by 2026-09-04 01:00."

8.3 Daily operation

scripts/run_daily_live_cycle.py is the schedulable batch half: fetch both feeds in the same cycle (so their windows align — fetching them separately once produced zero overlapping hours), upload to HDFS, enrich on YARN, hand the file to the stream's input directory. AirQualityStream is a separate always-on service; dropping the file there is the whole handoff.

9. Seven Defects Found Only by Running the Code

Each of these existed in code that had been written and, in several cases, described as complete. None surfaced until the code was executed against real data on the cluster. They are recorded because "the code is written" and "the phase is done" turned out to be different claims.

#	Where	What happened	How it was caught
1	01c merge	substring file-name join double-counted Mumbai and Noida (navi_mumbai.json contains mumbai.json)	row count 44,544 failed to reconcile with fetch count 44,160
2	AirQualityStream	referenced ts_unix one line before it existed - every batch aborted at history update	alerts printed BEFORE the failure, so it looked like a stall; only fires on a batch with history
3	AirQualityStream	VectorAssembler handleInvalid=skip silently dropped every live row (CAMS nh3 is always null)	16,560 rows in, 0 predicted, no error
4	paths.py	HDFS join produced hdfs:////namenode:9000/... (four slashes) - every HDFS-mode Python run would miss its own data	printed FEATURE_PATH inside the container

#	Where	What happened	How it was caught
5	Phase 4 write	DataNodes unresponsive under memory pressure; 4 nodes at replication 3 had no spare for the write pipeline	first full-scale run died after computing everything correctly
6	Spark image	numpy absent; PySpark ML aborts at import	training failed on first launch
7	Hive DDL	declared a column named timestamp that does not exist; Hive returned NULL for every time while counts looked right	first_alert / last_alert came back NULL

Defects 3 and 7 share a shape worth naming: both produced plausible-looking output with no error. Sixteen thousand rows in and zero out, with alerts and counts that looked fine; a Hive table returning correct counts and NULL timestamps. A silent wrong answer is worse than a crash, and the reconciliation habit — check the output count against an independently known number — is what caught them.

10. Hive Serving Layer

Three EXTERNAL tables (unified_airquality, clean_airquality, stream_alerts) and three views over the Parquet already in HDFS — no data copied, and DROP TABLE removes only metadata. Exposed through spark-sql and, via start-thriftserver.sh, over the HiveServer2 Thrift protocol on port 10000 so beeline and JDBC clients connect unchanged. All six demonstration queries return real results; the one to show is Delhi's seasonal cycle from measured values only:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
PM2.5	180.5	127.9	88.6	86.8	79.5	66.3	44.0	38.8	47.1	120.1	213.6	199.4

A 5.5× swing between the August monsoon low and the November stubble-burning peak. That is the physical reason 13 years of history is required to train the model, and why a live feed — however fresh — cannot replace it.

11. The Five V's

V	Evidence	State
Volume	154.6M rows, 33 GB, 1 NameNode + 4 DataNodes, real block distribution	demonstrated
Variety	flat CSV + nested self-describing JSON, read by native Spark readers into one schema; Hive queries span both	demonstrated
Veracity	4 dataset defects + 7 pipeline defects found and fixed; independent CPCB-vs-CAMS cross-validation reported honestly, including where it is weak	demonstrated
Velocity	Structured Streaming executed on the cluster: 10 s micro-batches, ~7 s latency, 1,130 alerts on genuinely live data	demonstrated
Value	GBT R2 0.8910 vs persistence 0.8587 on 2,074,569 real test rows, 12.18% RMSE reduction - an hour of warning before pollution arrives	demonstrated

12. Limitations, Disclosed

- 88.9% of structured rows are synthetic (statistically validated against real data, but not physical measurements). Model results are reported on real rows separately for that reason.
- Training used a seeded 10M-row sample of 74.5M available rows: a 15.7 GB RAM ceiling, not a pipeline limit. Evaluation was not sampled.
- CAMS is a reanalysis model, not a ground sensor; ammonia is null across the whole region.
- The semi-structured layers cover 230 of 241 cities and begin in August 2022; they do not reach back to 2010.
- Cross-validation agreement is moderate for particulates and poor for SO_2 and O_3.
- Small-cluster fragility is real: with 4 DataNodes at replication 3, a large derived write failed under memory pressure and was re-run at replication 2 with tuned timeouts. Replication 3 is kept for the raw layer, where fault tolerance is demonstrated.
- The dashboard (Phase 10) is not built.

13. Conclusion

The platform now does what its title says. Real CPCB history trains a forecaster that beats persistence by 12.18% on two million real test rows; real live atmospheric data flows through a Spark-on-YARN enrichment job into a Structured Streaming engine that fires an hour's warning above AQI 150; and everything it has produced is queryable through Hive. Along the way, seven defects that a code review would not have found were found by running the code — and each is documented with the number of rows it affected and the check that caught it.